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3 Complex Analysis 2024
 A/Prof Elena Berdysheva
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Deferred/Supplementary Final Exam
 January 2025



Instructions:

- This question paper consists of **8 pages** (including this one).
- You have 120 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 60 points = 100% (63 points available).

Problem	1	2	3	4	5	6	7	Σ	Mark
Maximum Points	10	8	6	9	12	12	6	60	100%
Result									

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Problem 1

(10 points)

Find the set of all $z \in \mathbb{C}$ where the function $f(z) = |z - i|^2$ is differentiable.

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Problem 2

(8 points)

Calculate the following integrals.

(a) $\int_{\gamma_1} \frac{1}{(z+i)^3} dz$, where $\gamma_1(t) = 2e^{it}$, $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

(b) $\int_{\gamma_2} \frac{1}{(z+i)^3} dz$, where $\gamma_2(t) = 2e^{it}$, $t \in [-\pi, \pi]$.

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Problem 3

(6 points)

Does there exist a holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$ with

$$f\left(\frac{1}{n}\right) = \frac{1}{n} \quad \text{for all } n \in \mathbb{N}, \quad f(-1) = 1?$$

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Problem 4

(9 points)

Use methods of complex analysis to calculate the integral $\int_0^{2\pi} \frac{1}{5 - 3 \cos \theta} d\theta$.

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Problem 5

(12 points)

Use methods of complex analysis to calculate P.V. $\int_{-\infty}^{\infty} \frac{1}{x^3 + 1} dx$.

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Problem 6

(12 points)

- (a) Find the radius of convergence of the power series $\sum_{n=0}^{\infty} \frac{(n!)^3}{(3n)!} z^{3n}$.
- (b) Determine the Laurent series of the function $f(z) = \frac{1}{9 - 4z^2}$ in the region $\{z \in \mathbb{C} : |z| > \frac{3}{2}\}$. Give an explicit formula for the coefficients a_n , $n \in \mathbb{Z}$, of the Laurent series.

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Problem 7

(6 points)

How many zeros does the polynomial $p(z) = z^{10} + z^5 + z^2 + 5$ have in the closed disc $\{z \in \mathbb{C} : |z| \leq 1\}$?